MIL-L-52496B(ME)
1 October 1980
SUPERSEDING
MIL-L-52496A(ME)
7 August 1968

# MILITARY SPECIFICATION

# LOADER, SCOOP TYPE, RUBBER-TIRED, DIESEL-ENGINE-DRIVEN, SECTIONALIZED AND NON-SECTIONALIZED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven, rubber-tired, scoop-type, front-end loader suitable for parachute delivery as a unit and for transport by helicopter when sectionalized.

1.2 Classification. The loaders shall be of the following types, as specified (see 6.2):

Type I - Non-Sectionalized (Air Transportable and Air-Droppable).
Type II - Sectionalized (Airborne-Transportable, Air-Droppable and Helicopter Transportable).

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

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| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-L-2104 | - Lubricating Oil, Internal Combustion Engine, Tactical Service. |

FSC 3805

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| MIL-L-2105 | - Lubricating Oil, Gear, Multipurpose. |
| MIL-T-3351 | - Tractor, Full-Tracked, Low Speed; Tractor, Wheeled, Agricultural; and Tractor, Wheeled, Industrial; and their Attachments, Packaging of. |
| MIL-M-7866 | - Molybdenum Disulfide, Technical, Lubrication Grade. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-L-46167 | - Lubricating Oil, Internal Combustion Engine, Arctic. |
| MIL-E-52798 | - Enamel, Alkyd, Camouflage. |

STANDARDS

MILITARY

| | |
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| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-209 | - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment. |
| MIL-STD-669 | - Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel. |
| MIL-STD-814 | - Requirements for Tiedown, Suspension and Extraction Provisions on Military Materiel for Air Delivery. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |
| MIL-STD-1410 | - Methods for Selection of Industrial Engines for End Item Application. |

MILITARY HANDBOOK

AIR FORCE SYSTEM COMMAND

| | |
|---|---|
| Design Handbook DH1-11 | - Air Transportability. |

(Copies of specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-H, Fort Belvoir, VA 22060.)



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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 General. The loader shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public for the same or similar application, interfacing with the same or similar components, except as specified herein, for at least one year prior to invitation for bids or request for proposal. Product improvements incorporated in the commercial production units in this one year period are acceptable. In the case of a recently introduced item, the contractor shall submit sufficient valid industrial operational/test data including all test reports for the item and its components, such as; engine, transmission and hydraulic system, field user reports, including any failures or problem areas, and corrective action taken to demonstrate acceptability of the item. When specific requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished. The loader shall be equipped with any optional equipment necessary to comply with requirements specified herein. Optional equipment is defined as equipment not standard with the loader but which has been furnished to the commercial customer, such as special features or allied equipment. The loader shall be equipped with all components necessary to enable the loader to function reliably and efficiently in sustained operation. The loader shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect on the date of invitation for bids or request for proposal.

3.2 Description. The loader shall be rubber-tired, diesel engine driven four wheel drive with rear axle oscillation and shall have an articulated frame steering system. The loader shall be equipped with a hydraulically operated, front mounted, scoop type, multipurpose bucket (see 3.18.1) and a quick coupler system (see 3.18.2).

3.2.1 Type I. The type I loader with ROPS/FOPS removed shall be capable of being loaded and rigged on an air delivery platform in accordance with MIL-STD-669, air transported and air delivered by parachutes.

3.2.2 Type II. The type II loader with ROPS/FOPS removed shall meet the requirements specified for type I (see 3.2.1) and shall also be capable of being sectionalized into two 2 sections for external air transport by helicopter. Sectionalization material may be removed for airdrop.

3.3 First article (preproduction models). The contractor shall furnish two loaders of each type for examination and demonstration within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce loaders that comply with the requirements of this specification. The examination and



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demonstrations shall be as specified in Section 4. Following successful completion of the examination and demonstration by the contractor, the Government shall perform a simulated airdrop demonstration on one of each type loader at a site selected by the Government. The examination and all demonstrations conducted by the contractor shall be subject to surveillance and approved by the Government. Acceptance of the preproduction models will be dependent on successful completion of the examination, contractor demonstrations, and the Government conducted simulated airdrop demonstration.

3.4 Production loaders. All loaders furnished under this specification shall be physically and mechanically identical to the First Article loaders approved by the Government.

3.5 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.6 Safety. All available standard or optional accessories shall be included to provide protection for maintenance and operating personnel from rotating or reciprocating parts and from all parts subject to high operational temperatures and so located as to be a hazard. Exhaust discharges shall be directed so as to not endanger personnel. All platforms and steps shall have anti-skid surfaces. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the loader can be set into motion. Contractor's standard grab irons shall be provided. A decal shall be furnished and shall be located in the area of the articulation point, warning of limited personnel clearance during frame articulation and placement of safety pin or bar for the loader operating and transport modes.

3.6.1 Sound level. The exterior sound level of the loader shall not exceed 88 dB(A) when measured in accordance with SAE J88. If the sound level in the area occupied by the operator exceeds 85 dB(A) when measured in accordance with SAE J919, a warning plate containing the following shall be permanently affixed to the loader in a conspicuous protected location.

"WARNING, EAR PROTECTION REQUIRED WHEN OPERATING THIS LOADER."

The plate shall have a yellow background with black lettering and shall be on corrosion-resistant material.

3.6.2 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The loader shall be equipped with a removable bolted down ROPS/FOPS canopy. The canopy shall be removable and replaceable without cutting or welding. The ROPS/FOPS shall conform to SAE J1040 and SAE J231, respectively. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. Quick disconnect fittings (electric, etc.) shall be provided as applicable to facilitate canopy removal. A slinging eye(s) shall be provided to facilitate removal of ROPS/FOPS.



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3.6.3 Fenders. Front and rear fenders shall be furnished. Rear fenders may be an integral part of the rear housing or structure.

3.7 Physical characteristics.

3.7.1 Weight limits. The type I and II loaders with the ROPS/FOPS structure removed but with all other attachments, accessories, coolant, lubricating and hydraulic systems filled to capacity and with not less than 25 percent or more than 75 percent of fuel shall weigh not more than 30,600 pounds. The type II loader when sectionalized into two (2) sections shall weigh not more than 16,000 pounds per section.

3.7.2 Dimension limits. The overall width (excluding the bucket), and length of the type I and II loaders shall not exceed 105 inches and 336 inches respectively. The bucket width shall not exceed 108 inches. In order to conform to a height limit of 100 inches when rigged for airdrop, the reducible height of the loader should be not more than 91 inches. No major component removal, except ROPS/FOPS canopy, shall be required to obtain the reduced height. If the height exceeds 91 inches, the contractor must demonstrate prior to step I proposal opening date that his loader can be rigged so as to conform to the 100 inch limit, and withstand the forces encountered in the simulated airdrop impact test (see 4.5.2.11) without damage or permanent deformation.

3.8 Sectionalization. The type II loader less ROPS/FOPS shall be capable of being sectionalized into two (2) sections for helicopter transport. The weight of either section shall not exceed 16,000 pounds. The powered section shall be driveable and steerable under its own power for a distance of 50 feet over relatively flat unimproved areas. Reassembly time with tools furnished with the loader under field conditions without the assistance of other powered equipment shall not exceed four (4) hours. Cutting and welding are not permitted. Components and accessories used in sectionalization and reassembly shall be stored on the loader or in a demountable container attached to the loader but shall not interfere with the operation of the loader.

3.9 Performance. The loader and attachments shall withstand the usage encountered in military operations such as transport by helipcoter, air delivery, loading trucks from stockpiles, stockpiling material, and excavating undisturbed and compacted soil without breakage, malfunction or permanent deformation. The loader shall also be used as a clamshell to handle irregularly shaped objects, as a dozer for general bulldozer work and as a scraper.

3.9.1 Environmental conditions. Without the use of any external means, equipped with only standard or optional equipment, the engine shall start within five minutes and the loader shall operate under full-load conditions within fifteen minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be integral with the loader and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The loader shall be capable of operating in hot, humid tropical environments and dry, dusty environments.



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3.9.2 Travel speed. The empty loader shall be capable of a forward travel speed of not less than 20 mph on level pavement.

3.9.3 Stability. The loader shall negotiate 27 percent side slopes and 58 percent longitudinal slopes without any wheel of the loader losing contact with the ground. All parts requiring lubrication shall be lubricated when the loader is operating under these conditions.

3.9.4 Fording. The loader shall start and operate during and after fording operations in fresh water to any depth up to 20 inches including wave action without damage and without leakage of water into reservoirs containing lubricants.

3.9.5 Operating load. The loader shall have a SAE rated operating load of 8,000 pounds (reference SAE J818).

3.9.6 Tipping load. The loader shall have a turn tipping load, SAE rating, of not less than 16,000 pounds (reference SAE J732).

3.9.7 Lift capacity. The loader shall have an SAE lift capacity at the carry position of not less than 20,000 pounds. The loader shall have an SAE lift capacity to maximum height of not less than 10,000 pounds (reference SAE J732).

3.9.8 Breakout force. The loader shall have an SAE breakout force of not less than 17,500 pounds (reference SAE J732).

3.9.9 Clam closure force. The closure force at the lip of the closed clam shall be not less than 7,000 pounds.

3.9.10 Clearance circle. The loader shall have an SAE clearance circle diameter of not more than 44 feet (reference SAE J732).

3.10 Engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The diesel engine shall be the engine furnished commercially including all systems, components, accessories, and auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow.

3.10.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816, shall be not less than 130 horsepower.

3.10.2 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.9.1 shall be provided.

3.10.3 Fuel tank. The fuel tank shall be of a type normally furnished as standard equipment with a capacity for 10 hours of continuous normal operation. The tank shall be equipped with a means for draining water and sediment from the bottom of the tank.



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3.10.4 Engine side panels. The loader shall be equipped with removable engine side panels with quick-disconnect latches or horizontally or vertically hinged engine compartment doors.

3.11 Drive train.

3.11.1 Transmission. A full powershift transmission shall be furnished. The transmission shall provide not less than three forward speeds and three reverse speed; or when a dual range, twin turbine torque converter is furnished not less than two forward gears and one reverse gear resulting in four forward speeds and two reverse speeds. The loader shall be equipped with a transmission disconnect that automatically disconnects the drive train when the service brakes are applied. The loader shall also have the capability of having the service brake applied without disconnecting the drive train. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the loader can be set in motion.

3.11.2 Torque converter. The loader shall be equipped with a torque converter. A torque converter oil cooler shall be furnished.

3.11.3 Drive axles. The front and rear drive shall have differentials and planetary final drives. A limited slip differential or a hydraulic differential lock shall be furnished in the rear axle or torque transferring differentials of the torque bias type, in the front and rear axles or front axle only.

3.12 Tires. Tires shall be of the traction type tread (L-2), nylon, tubeless, with not less than a 12 ply rating. Tires shall be of the wide base type with a size of 20.5 x 25. All tires shall be of the same size, type, and ply rating.

3.13 Steering and braking.

3.13.1 Steering system. The loader shall have an articulated frame type steering system with a minimum of 30 degrees of articulation in each direction. The steering system shall provide full control of the loader under all normal operating conditions and shall provide emergency steering conforming to SAE J53 without engine power.

3.13.2 Braking system. A foot controlled service braking system shall be provided with disc brakes acting on all four wheels. A parking system and an emergency stopping system shall be provided. The braking system shall comply with SAE J1152.

3.14 Operator's compartment.

3.14.1 Seat. The contractors standard commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone and all instruments and gages, mounted on the instrument panel, when seated with seat belt fastened and properly adjusted.



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3.14.2 Instruments and indicators. The following instruments and indicators shall be provided:

(a) Engine oil pressure gage.
(b) Engine coolant temperature gage.
(c) Torque converter oil temperature gage.
(d) Ammeter or voltmeter.
(e) Engine hourmeter or service meter.
(f) Air pressure gage (when equipped with air system).
(g) Low air pressure warning buzzer (when equipped with air system).
(h) Fuel level indicator (dipstick or gage).
(i) Hydraulic oil level indicator (dipstick, sight tube, or gage).
(j) Air cleaner restriction indicator.
(k) Speedometer or tachometer (if offered).
(l) Odometer (if offered).

Additional instruments and indicators standard with the contractor's commercial practice, which contribute to the safety and reliability of loader operation, shall be furnished.

3.14.3 Controls. All controls for the loader shall be located in the operator's compartment and shall be within reach of the operator in his normal operating position.

3.15 Hydraulic system. The loader shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all attachments and related components. An adjustable automatic boom kickout and an adjustable automatic bucket positioner shall be furnished.

3.16 Electrical system. The loader shall be equipped with the contractor's standard or optional electrical system which meets the requirements specified herein. A 24 volt starting system shall be furnished if available as standard or as an option.

3.16.1 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.9.1.

3.16.2 Alternator. The contractor's high output option alternator shall be furnished. The rated output shall be not less than 50 amps.

3.16.3 Lighting. The following lights with associated circuits, circuit breakers, or other protective devices and switches shall be provided if available as standard equipment or as an option:

(a) Four (4) front work/travel lights.
(b) Two (2) combination stop and taillights.
(c) One (1) rear work light.
(d) Turn signals with emergency flashers (if offered).
(e) Instrument panel lighting.

3.16.4 Electromagnetic radiation. The loader shall comply with the electromagnetic radiation limits of SAE J551.



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3.16.5 Horn. A conventional operator controlled horn shall be furnished.

3.16.6 Backup alarm. A backup alarm conforming to SAE J994 shall be furnished.

3.17 Towing device. The loader shall be equipped with a fixed type drawbar including a drawbar pin and a positive pin retaining means.

3.18 Attachments.

3.18.1 Multipurpose bucket. The loader shall be furnished with a multipurpose type bucket. The SAE nominal heaped bucket capacity rating shall be 2-1/2 cubic yards when rated in accordance with SAE J742. The bucket shall be equipped with not less than eight bolt-on teeth. The bucket width shall be at least 1 inch greater than the loader without bucket. The bucket shall operate as a front loader, clamshell, dozer, and scraper.

3.18.2 Quick coupler system. The loader shall be equipped with a quick coupler system than allows rapid exchange of loader-mounted front end attachments. Actuation method of the locking pins is optional. Pin actuation shall be controlled by the operator from the operator's compartment.

3.18.3 Counterweight. Removable rear counterweight may be furnished on the loader to improve stability and to meet the requirements of this specification. The counterweight furnished shall be the same as that furnished by the contractor on his standard commercial loader when equipped with a 2-1/2 cubic yard multipurpose bucket. Tire ballast is not acceptable and shall not be furnished.

3.19 Transportability. The loader shall be capable of being transported by surface, rail, ocean-going carriers, and C130, C141, and C5A aircraft. The loader shall withstand the impact forces encountered in air and rail shipments without damage or permanent deformation. The loader shall conform to MIL-STD-669 for airdrop from C130 aircraft and shall withstand stresses imposed when tested in accordance with 4.5.2.11 without breakage or permanent deformation of any structural member or component, or loss of ability of the loader to perform all of its intended functions. When rigged for airdrop in accordance with MIL-STD-669, the loader shall meet the established tipoff curves for C130 and C141 aircraft as defined in AFSC Design Handbook DH1-11. The loader shall be equipped with slinging provisions and shall be capable of being lifted without damage using slings. The loader shall be equipped with tiedown provisions for transport by the above modes. Suspension and extraction parachute provisions shall be provided for parachute delivery. Complete diagrams and instructions for lifting the loader and tying down the loader on freight cars and truck trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport. The contractor shall provide sufficient data to the Government with the step I proposal to confirm that the loader being supplied can be loaded and transported in the C130, C141, and C5A aircraft. A steering frame lock in accordance with SAE J276 shall be provided. Removal of ROPS and other components to reduce overall height or width shall not prevent the loader from driving onto military transport equipment under its own power.



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3.19.1 Parachute suspension provisions. Four integral suspension provisions conforming to MIL-STD-814 shall be furnished on the loader. The parachute suspension provision shall withstand the stresses in the amount and direction of pull specified without weld failure or permanent deformation of the loader or the provision.

3.19.2 Slinging provisions. The loader shall be provided with four slinging provisions conforming to MIL-STD-209. In addition each section of the type II loader shall be provided with four (4) slinging provisions conforming to MIL-STD-209. The provisions shall enable the loader to be lifted in the normal traveling or operating position. Provisions shall be positioned so that the lifting strain shall be in line with the longitudinal axis of the eye of the provisions. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the loader or the provision. Parachute suspension provisions may be used as slinging provisions if they meet the requirements of MIL-STD-209.

3.19.3 Tiedown provisions. The loader shall be provided with integral provisions to permit tiedown of the loader to the floor or deck of the transportation medium or the airdrop platform. The tiedown provisions shall conform to MIL-STD-209 and MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the loader or the provision.

3.19.4 Extraction parachute provision. A single fitting shall be provided on the loader for attachment of the parachute extraction system. Except as specified herein, the fitting shall conform to MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the loader or the provision. In lieu of the limit load specified in MIL-STD-814, the limit load of the provision shall be 56,580 pounds. The design ultimate strength of welded sections shall be four (4) times the limit load.

3.20 Identification plates and marking. Each load shall be furnished with an identification plate showing the contractor's model number, NSN, USA Registration number, date of manufacture, and date of delivery. The identification plate shall be permanently affixed to the loader in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly. The loader and each attachment shall be identified in accordance with the contractor's standard method.

3.21 Instruction plates. The loader shall be equipped with instruction plates including warnings and cautions suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the loader. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. Instructions for disassembly of the type II loader into two sections and reassembly shall be provided on cards approximately 8-1/2 inches by 11 inches which shall be encased in laminated plastic. One set shall be furnished with each type II loader.



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3.22 Shipping data plates. Shipping data plates for each loader shall be furnished and shall show the silhoutte of the loader and each section (type II loader only), indicating the location and capacity of slinging provisions, tiedown provisions, extraction provisions, suspension provisions, and the center of balance and shipping weight. Plates shall conform to MIL-P-514 or *contractor's standard plates subject to the approval* of the contracting officer. The plates shall be attached to the left side of the loader and shall show the locations as viewed from the left side. The plates on the sections, (type II loader only) shall be located in conspicuous protected locations. All plates shall be permanently attached.

3.23 Treatment and painting. The portions of the loader normally painted shall be cleaned, treated, primed and painted in accordance with the contractor's standard commercial practice. The finishing coat shall be in accordance with MIL-E-52798, forest green color.

3.24 Stencling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown, slinging, suspension and extraction provisions, tire pressure, and the center of gravity for the type I and type II loaders, and the sectionalization storage areas on the type II loader.

3.25 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each loader shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924 and MIL-G-23827 as applicable.

3.26 Technical publications. Such technical publications as are specified (see 6.2) shall be furnished.

3.27 Special tools. Any nonstandard or special tools required for equipment operation and operator maintenance of the loader shall be furnished with the loader.

3.28 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.29 Technical information package. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2) depicting the loader and its components exactly as it will be furnished. The Technical Information Package shall be submitted with all bids and proposals.

3.30 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.3):



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(a) Container NSN 7520-00-559-9618 complete with maintenance forms. (The contractor shall furnish the fasteners and install the containers on the loaders.)
(b) Supplemental operating, maintenance and repair parts instructions (SOMARPI).

3.31 Basic issue items. When specified (see 6.2), each item in the Basic Issue Item List shall be furnished with each loader.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection. The first article (preproduction model) loader (see 6.2), furnished by the contractor shall be examined (see 4.3.1), and demonstrated (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and demonstration of the loader the following shall be performed:

(a) Service the loader with the lubricants specified.
(b) Break in the loader as prescribed by the contractor.

4.3.1 Preproducton examination. Prior to the preproduction demonstration, the loader shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection. Prior to the preproduction demonstration, the loader shall be examined and compared with this specification, with the contractor's published brochures, with commercial manuals, and with the completed Technical Information Package. Any discrepancy between the specification, brochures, manuals, or Technical Information Package and the preproduction model shall be cause for rejection. Prior to the preproduction demonstration, the certifications, and certified test reports required by 4.5.2.12, and 4.5.2.13 shall be examined. Incomplete, inaccurate, or otherwise unacceptable certifications, or certified test reports shall be cause for rejection.



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4.3.2 Preproduction demonstration. Upon successful completion of the examination specified in 4.3.1, the preproduction model loaders shall be subjected to the demonstrations marked "X" in column 1 of table II. Failure of any demonstration or any demonstration reports shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the demonstration specified in 4.4.2, each loader shall be examined for the defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual demonstration. Each loader shall be demonstrated as specified in column 2 of table II. Failure of any demonstration shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table I.

4.5.2 Demonstrations. Demonstrations shall be conducted in accordance with table II in the sequence listed.



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TABLE I. Examination schedule.

| Preproduction model(s) | Individual | Defect | | Requirements Paragraph |
|---|---|---|---|---|
| 1 | 2 | 3 | | 4 |
| X | X | 101. | Safety items not as specified. | 3.6 |
| X | X | 102. | Missing warning decal (if required). | 3.6.1 |
| X | X | 103. | ROPS/FOPS canopy mounting not as specified. | 3.6.2 |
| X | X | 104. | Fenders not as specified. | 3.6.3 |
| X | - | 105. | Dimension limits not as specified. | 3.7.2 |
| X | X | 106. | Cold start aid not as specified. | 3.9.1 |
| X | X | 107. | Exhaust system not as specified. | 3.10 |
| X | X | 108. | Fuel tank draining provisions not as specified. | 3.10.3 |
| X | X | 109. | Engine side panels not furnished. | 3.10.4 |
| X | X | 110. | Transmission not as specified. | 3.11.1 |
| X | X | 111. | Torque converter not as specified. | 3.11.2 |
| X | X | 112. | Drive axles not as specified. | 3.11.3 |
| X | X | 113. | Tires not as specified. | 3.12 |
| X | X | 114. | Steering systems not as specified. | 3.13.1 |
| X | X | 115. | Braking system not as specified. | 3.13.2 |
| X | X | 116. | Seat not as specified. | 3.14.1 |
| X | X | 117. | Instruments and indicators not as specified. | 3.14.2 |
| X | X | 118. | Controls not as specified. | 3.14.3 |
| X | X | 119. | Hydraulic system not as specified. | 3.15 |
| X | X | 120. | Alternator not as specified. | 3.16.2 |
| X | X | 121. | Lighting not as specified. | 3.16.3 |
| X | X | 122. | Horn not as specified. | 3.16.5 |
| X | X | 123. | Backup alarm not as specified. | 3.16.6 |
| X | X | 124. | Towing device not as specified. | 3.17 |
| X | X | 125. | Multipurpose bucket not as specified. | 3.18.1 |
| X | X | 126. | Quick coupler system not as specified. | 3.18.2 |
| X | X | 127. | Counterweight not as specified. | 3.18.3 |
| X | X | 128. | Steering frame lock not as specified. | 3.19 |
| X | X | 129. | Identification plates and marking not as specified. | 3.20 |
| X | X | 130. | Instruction plates not as specified. | 3.21 |
| X | X | 131. | Shipping data plates not as specified. | 3.22 |
| X | X | 132. | Treatment and painting not as specified. | 3.23 |
| X | X | 133. | Stenciling instruction not as specified. | 3.24 |



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TABLE I. Examination schedule. (Cont'd)

| Preproduction model(s) | Individual | Defect | Requirements Paragraph |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| X | X | 134. Lubrication not as specified. | 3.25 |
| X | X | 135. Technical publications not as specified. | 3.26 |
| X | X | 136. Special tools not as specified. | 3.27 |
| X | X | 137. Toolbox not as specified. | 3.28 |
| X | - | 138. Technical information package incomplete or not up to date. | 3.29 |
| X | X | 139. Government-furnished equipment not installed. | 3.30 |
| X | X | 140. Parts or components missing. | Section 3 |
| X | X | 141. Assembly incomplete or incorrect. | Section 3 |
| X | X | 142. Cracked or broken parts or assemblies. | Section 3 |
| X | X | 143. Leaks in coolant, fuel or oil lines. | Section 3 |
| X | X | 144. Permanent deformation of any part or assembly. | Section 3 |
| X | X | 145. Material not of same quality as used on preproduction model. | Section 3 |
| X | X | 146. Loader not as specified. | Section 3 |

TABLE II. Demonstration schedule.

| Preproduction model(s) | | Individual | Demonstration | Demonstration Paragraph | Requirements Paragraph |
|---|---|---|---|---|---|
| 1 | | 2 | 3 | 4 | 5 |
| Type I | Type II | | | | |
| X | X | X | Production run-in | 4.5.2.1 | 3.25 |
| - | X | - | Sound level | 4.5.2.2 | 3.6.1 |
| X | X | - | Weight | 4.5.2.3 | 3.7.1 |
| X | X | - | ROPS removal | 4.5.2.4 | 3.6.2, 3.19 |
| - | X | - | Sectionalization | 4.5.2.5 | 3.8 |
| X | X | - | Parachute suspension provisions | 4.5.2.6 | 3.19.1 |
| X | X | - | Slinging provisions | 4.5.2.7 | 3.19.2 |
| X | X | - | Tiedown provisions | 4.5.2.8 | 3.19.3 |
| X | X | - | Extraction parachute provisions | 4.5.2.9 | 3.19.4 |
| - | X | - | Construction tasks | 4.5.2.10 | 3.9, 3.10.3 |
| X | X | - | Airdrop | 4.5.2.11 | 3.19 |



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4.5.2.1 Production run-in. Start and operate each loader for not less than 1/2 hour to assure operation of all components and to allow for minor adjustments. Start and stop the engine not less than five times. Operate all bucket controls for five minutes. Hold each hydraulic control in the actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the loader at least two times through a figure eight steering course at the maximum safe speed. Make not less than five abrupt panic stops from a speed not less than 20 mph using the service brakes. Release and reapply the parking brakes not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the loader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that loaders previously accepted are similarly defective unless evidence, satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.5.2.2 Sound level. The contractor shall perform the sound level tests in accordance with SAE J88 and SAE J919. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.6.1 shall constitute failure of this demonstration.

4.5.2.3 Weight. The weight of the loader and loader sections shall be measured with a scale calibrated within six (6) months of the demonstrations. Inability to meet the requirements of 3.7.1 shall constitute failure of this demonstration.

4.5.2.4 ROPS/FOPS removal and replacement. The removal and replacement of the ROPS/FOPS canopy shall be demonstrated following procedures in the contractor's manual. Permanent deformation or damage to any part of the ROPS/FOPS canopy or its mounts, or inability to meet the requirements of 3.6.2 and 3.19 shall constitute failure of this demonstration.

4.5.2.5 Sectionalization. The type II loader less ROPS/FOPS shall be demonstrated for compliance to sectionalization requirements of 3.8. The demonstration shall be conducted on a relatively flat unimproved area of sufficient size for disassembly, maneuvering, and reassembly of the loader. Using hand tools furnished with the loader (no power tools allowed), separate the loader into two sections. Drive the powered section under its own power for a distance of 50 feet from the remaining section and steer the powered section at least 45 degrees to the longitudinal axis of the remaining section. Drive and steer the powered section to join with the remaining section and reassemble the two sections into a composite unit using the above hand tools only. Record the time required for the maneuvering and reassembly operation. No chains, cables, hoists, blocking, or other vehicles shall be used to assist in the disassembly, maneuvering, and reassembly operations. Nonconformance to 3.8 shall constitute failure of this demonstration. The sectionalization demonstration will be compared to the disassembly and assembly instructions specified in 3.21.

4.5.2.6 Parachute suspension provisions. The parachute suspension provisions shall be demonstrated to prove conformance to 3.19.1. Apply a force of 1.5 times the limit load as specified in MIL-STD-814 to each of the parachute



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suspension provisions. Weld failure or permanent deformation of the loader or suspension provisions shall constitute failure of this demonstration.

4.5.2.7 Slinging provisions. The slinging provisions shall be demonstrated to prove conformance to 3.19.2. Nonconformance to 3.19.2, weld failure, or permanent deformation of the loader or the slinging provisions shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

(a) Lift the loader and hold in suspension in its normal travel prosition using slings that converge not more than 24 feet above the lowest extremity of the loader. Spreader bars are permitted. Determine the force and angles of application for each provisions. Measure the clearance between each sling and the loader.
(b) Restrain the loader by anchoring the main frame and subject each slinging provisions to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for a minimum of 90 seconds.
(c) Repeat (a) and (b) above for each section of the type II loader.

4.5.2.8 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.19.3. Apply a force equal to the limit load as specified in MIL-STD-814 to each of the tiedown provisions. Weld failure or permanent deformation of the loader or the tiedown provisions shall constitute failure of this demonstration.

4.5.2.9 Extraction parachute provision. The extraction parachute provision shall be demonstrated to prove conformance to 3.19.4. Apply a force equal to 1.5 times the limit load specified in 3.19.4 to the provision. Weld failure or permanent deformation of the loader or the extraction parachute provision shall constitute failure of this demonstration.

4.5.2.10 Construction tasks. The loader shall be demonstrated as follows for a total of 40 hours to prove conformance to 3.9 and 3.10.3.

(a) Using bank run gravel or building sand demonstrate for 15 hours loading (from stockpiles) dump trucks having a sideboard height of not less than 80 inches above ground level. Approximately one-half of this operation shall be conducted with the load bottom dumped.
(b) Using bank run gravel or building sand demonstrate for 15 hours the loader's ability to move stockpiles with capacity loads a distance of not less than 100 feet. Approximately one-half of this operation shall be conducted with load bottom dumped.
(c) Demonstrate for 5 hours the loader's ability to excavate a pit in undisturbed soil using the bucket closed.
(d) Demonstrate for 5 hours the loader's ability to excavate a pit in undisturbed soil using the open bucket as a dozer.

Nonconformance to 3.9 and 3.10.3 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this demonstration.



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4.5.2.11 Airdrop. The loader shall be subjected to simulated airdrop impact demonstration in accordance with MIL-STD-669 to prove conformance to 3.19. This demonstration will be conducted by the Government at a Government facility; however, acceptance of the preproduction models will be dependent on the successful completion of this demonstration. The demonstration shall consist of the following:

(a) Rig the machine in accordance with MIL-STD-669 and confirm rigged weight and dimensions conform to the requirements of AFSC Design Handbook DH1-11.
(b) Drop test the machine in accordance with MIL-STD-669.
(c) After the first drop, the machine shall be de-rigged, examined, started, and driven forward and backward a few feet, and all controls shall be actuated.
(d) The machine shall then be sectionalized and reassembled in accordance with the contractor's instruction to ascertain that no damage was done to the sectionalization features by the test drop. (This step is ommitted for the type I machine).
(e) Rerig and drop test the machine a second time in accordance with MIL-STD-669.
(f) After de-rigging, the loader shall be demonstrated for eight 8 hours performing 20 percent of each of the construction tasks in 4.5.2.10.

Failure to meet the weight and dimensional requirements of AFSC Design Handbook DH1-11, weld failure, leakage, permanent deformation of any structural member or component, inability to reassemble the components removed for airdrop, inability to disassemble and reassemble the type II loader, or inability of the loader to perform its intended functions shall constitute failure of this demonstration.

4.5.2.12 Certification. The contractor shall certify in writing to the contracting officer that the loader complies with the starting and operating temperature requirements of 3.9.1. The stability requirement of 3.9.3 the fording requirement of 3.9.4, the operating load requirement of 3.9.5, the tipping load requirement of 3.9.6, the lift capacity requirement of 3.9.7, the breakout force requirement of 3.9.8, the clam closure force requirement of 3.9.9, the clearance circle requirement of 3.9.10, the engine requirements of 3.10, the horsepower requirement of 3.10.1, the steering requirement of 3.13.1, the braking requirement of 3.13.2 and the rail shipment requirement of 3.19. If certification is not supplied, these requirements shall be demonstrated. Nonconformance shall constitute failure of this demonstration.

4.5.2.13 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS/FOPS requirements of 3.6.2, the travel speed requirement of 3.9.2 and electromagnetic radiation of 3.16.4. If certified test reports are not available, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with MIL-T-3351 (for level A or B) and MIL-STD-1188 (for commercial).



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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each complete loader, components and attachments shall be preserved in accordance with the level A preservation and packaging requirements of MIL-T-3351.

5.1.2 Level B. Each complete loader, components and attachments shall be preserved in accordance with level B preservation and packaging requirements of MIL-T-3351.

5.1.3 Commercial. Each complete loader and all items issue with the loader shall be suitably preserved and packed in accordance with MIL-STD-1188 for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservation coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the loader in operation.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

5.2.2 Level B. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

5.2.3 Commercial. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188. Commercial packing must protect items against physical and environmental damage during shipment, handling and storage.

5.3 Marking.

5.3.1 Military. Marking for level A or B protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.



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## 6. NOTES

6.1 Intended use. The loader is intended to be used in airborne and air assault operations primarily for use as a bucket loader for long-range stockpile work, excavating, and general utility work, and as an expedient replacement for small cranes and shovels.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and data of this specification.
(b) Type loader required (see 1.2).
(c) Time frame for submission of preproduction model (see 3.3).
(d) Technical publications required (see 3.26).
(e) Time frame for submission of Technical Information Package (see 3.29).
(f) When Basic Issue Items are to be furnished (see 3.31).
(g) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.30.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

Custodian:
Army - ME

Preparing activity:
Army - ME

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-L-52496B(ME) Loader, Scoop Type, Rubber-Tired, DED, Sectionalized and Non-sectionalized

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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